

Digitaler Materialpass

ID DMP-METAL-003 - Version 1.0.0

Ausstellungsdatum 2025-05-15 - Zertifikatstyp EN 10204 3.1

Hersteller	Kunde	Geschäftsvorgang	
ACME Metal Works GmbH Industrial Park 123 52066 Aachen DE quality@acme-metal.example.com	Precision Aerospace Inc. Aviation Boulevard 789 Mountain View 94043 US materials@precision-aero.example.com	Bestellung	PO-34567 · 500 kg Pos. 3 · 2025-04-25
		Lieferung	DN-89012 · 500 kg Pos. 1 · 2025-05-14 Interne Bestellnummer: SO-67890

Produkt

Produktname	Aluminum Alloy 7075-T6	Name (AA)	7075-T6
Chargen-ID	H-43210-01	Nummer (UNS)	A97075
Oberflächenzustand	Rolled	Form	Plate - Width 1000 mm - Thickness 10 mm - Length 2000 mm
Gewicht	500 kg	Produktnorm	AMS 4045 (2023)
Produktionsdatum	2025-05-12		
Ursprungsland	DE		

Chemische Analyse

Schmelzenummer H-43210 - Schmelzverfahren VAR - Gießverfahren VacuumCasting - Gießdatum 2025-05-10 - Probenort Ladle

Symbol	Al	Zn	Mg	Cu	Cr	F1
Einheit	%	%	%	%	%	
Min		5.1	2.1	1.2	0.18	0.18
Max		6.1	2.9	2.0	0.28	0.28
Ist-Wert	89.7	5.6	2.4	1.5	0.22	0.22

F1 = C+Mn/6+(Cr+Mo+V)/5+(Ni+Cu)/15: 0.22

Mechanische Eigenschaften

Tensile Strength ¹											
	Symbol	1	2	3	Durchschnitt	Min	Max	Std.-Abw.	Spez. Min	Methode	Status
Wert [MPa]	Rm	570	572	574	572	570	574	2	530	ASTM E8	✓
0.2% Yield Strength ¹											
	Symbol	1	2	3	Durchschnitt	Min	Max	Std.-Abw.	Spez. Min	Methode	Status
Wert [MPa]	Rp0.2	503	505	507	505	503	507	2	480	ASTM E8	✓
Elongation ¹											
	Symbol	1	2	3	Durchschnitt	Min	Max	Spez. Min	Methode	Status	
Wert [%]	A	10.8	11.2	11	11	10.8	11.2	10	ASTM E8	✓	

¹ 3 specimens tested

Physikalische Eigenschaften

Eigenschaft	Symbol	Ist-Wert	Soll/Min	Maximum	Methode	Status
Density	ρ	2.81 g/cm³	2.81 g/cm³		ASTM B311	✓
Coefficient of Thermal Expansion Temperature range: 20-100°C	α	23.4 10 ⁻⁶ /K	23.5 10 ⁻⁶ /K		ASTM E228	✓
Thermal Conductivity At 25°C	λ	130 W/(m·K)	120 W/(m·K)		ASTM E1461	✓
Specific Heat Capacity At 25°C	cp	862 J/(kg·K)	860 J/(kg·K)		ASTM E1269	✓
Electrical Resistivity At 20°C	ρe	0.0538 μΩ·m		0.0550 μΩ·m	ASTM B193	✓
Poisson's Ratio	ν	0.33	0.33		ASTM E132	✓
Melting Range	Tm	477 - 635 °C	475 - 635 °C		ASTM E1142	✓
Relative Magnetic Permeability	μr	1.00002		1.00010	ASTM A342	✓
Surface Roughness	Ra	0.8 μm		1.6 μm	ISO 4287	✓
Emissivity At 25°C	ε	0.09		0.11	ASTM E408	✓
Surface Tension	γ	0.875 N/m	0.870 N/m		ASTM D971	✓
Diffusion Coefficient	D	0.0000000023 m²/s	0.0000000022 m²/s		ASTM E1559	✓
Purity Plate II, inspected area approx. 200 mm²					ISO 4967 method B	

Inclusion type	A fine	A thick	B fine	B thick	C fine	C thick	D fine	D thick	DS
Max	2.5	2	2	1	1.5	1	1.5	1	1.5
Wert	1	1	1.5	0.5	0.5	0	1	0.5	0.5

Validierung

We hereby certify that the material described above has been manufactured and tested in accordance with AMS 4045 and the specified test methods. All results are within the specified limits.

Validiert von **Elsa Müller**, Materials Engineer, Quality Assurance - 2025-05-15